

VitalSync

User Manual

Version 1.0 | Android 10+ | June 2025

github.com/sohiaking/VitalSync

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1. Getting Started

1.1 System Requirements

- Android 10 (API 29) or higher
- ~25 MB free storage space
- Internet connection required only for Walk Tracker map tiles

1.2 Installation

1**Download the APK**

Open github.com/sohiaking/VitalSync on your Android device and tap the Download APK button.

2**Allow Installation**

When prompted, tap 'Allow from this source'. You can find this in Settings → Apps → Special app access → Install unknown apps.

3**Install**

Open the downloaded VitalSync.apk file from your notifications or Downloads folder and tap Install.

4**Launch**

Once installed, open VitalSync from your home screen. No account or sign-in is required.

5**Grant Permissions**

On first launch, grant the requested permissions (steps, location, notifications) for full functionality.

2. Home Dashboard

The Home Dashboard is your daily health overview. It loads automatically on app launch and displays all key metrics in one place.

2.1 Dashboard Components

- **Step Progress Ring** — circular gauge showing steps completed vs. the 10,000-step daily goal
- **Distance** — total kilometers walked today
- **Calories Burned** — estimated kcal based on step count and body weight
- **Hydration Card** — glasses of water consumed out of daily goal (default 8)
- **Sleep Summary** — last logged sleep with duration; tap to log new sleep
- **Daily Progress Section** — scrollable cards for all modules
- **SOS Button** — red emergency button always visible top-right

2.2 Navigation Bar

The bottom navigation bar contains five tabs: Home, Activity, Nutrition, Gym, and Health. Tap any tab to switch modules instantly. The active tab is highlighted in green.

3. Activity Tracker

3.1 Step Counter

VitalSync uses your phone's built-in hardware step sensor (TYPE_STEP_COUNTER) for accurate step detection. Steps are counted automatically whenever the app is running in the background.

- The large circular display shows steps for today
- Below it: Distance (km), Calories (kcal), and Goal % are updated in real time
- Use the + / – buttons to manually adjust step count (e.g. for treadmill use)

3.2 Walk Tracker

Walk Tracker records your GPS route in real time and displays it on a Google Maps view.

- Tap Track Activity on the Home Dashboard to begin a walk session
- Your route is drawn as a green line as you move
- Walk sessions are saved locally with date, distance, and duration
- Ensure Location permission is granted for this feature to work

4. Nutrition

4.1 Hydration Tracker

The hydration section displays 8 glass icons. Tap each glass icon to mark it as consumed (each glass = 250ml). Tap again to un-mark. The progress bar at the bottom shows total ml consumed vs. the 2,000ml daily goal. Use + / – buttons to change the number of glasses.

4.2 Calorie & Macro Tracking

The Today's Calories section shows total calories consumed. On the right, Protein, Carbs, and Fat are displayed in grams. A progress bar shows intake vs. the 2,000 kcal daily target.

4.3 Meal Log

- Tap + Add Meal to log a new meal
- Enter meal name, calories, and optional macro breakdown (protein/carbs/fat)
- All meals are listed below with timestamp and calorie count
- Swipe left on a meal entry to delete it

5. Gym Tracker

5.1 Starting a Workout Session

The Gym Tracker starts a new session automatically when you open the module. An elapsed timer begins counting from 00:00.

5.2 Adding Exercises

- Tap any Quick Add button (Bench Press, Squat, Deadlift, Pull-ups) for instant add
- Tap + Add Exercise to search and add any custom exercise
- For each exercise, enter sets, reps, and optional weight (kg/lbs)

5.3 Rest Timer

Tap 30s, 60s, 90s, or 120s to start a countdown rest timer between sets. A notification will alert you when rest time is complete.

5.4 Finishing a Session

Tap the red Finish button to end the session. The workout is saved with total duration, all exercises, sets and reps. View past sessions in the workout history below.

6. Health Module

6.1 Sleep Tracker

To log sleep:

1. Tap + Log Sleep in the Sleep Tracker section
2. Select your bedtime (e.g. 23:00) and wake time (e.g. 07:00)
3. Rate your sleep quality (1–5 scale)
4. Tap Save — the entry appears with duration and quality emoji

The 7-day bar chart shows your sleep duration per night. The green band marks the recommended 7–9h zone. Statistics below show avg sleep, avg quality score, and sessions count.

6.2 BMI Calculator

- Enter your weight (kg) and height (cm) in the input fields
- BMI is calculated instantly and displayed with a color-coded result
- Color scale: Blue = Underweight | Green = Normal | Yellow = Overweight | Red = Obese
- Your last entered values are saved automatically

7. SOS & Fall Detection

7.1 Setting Up Emergency Contacts

Before using SOS, add at least one emergency contact in the Health module. Tap Add Emergency Contact, enter the name and phone number, and save.

7.2 Fall Detection

VitalSync continuously monitors the accelerometer in the background. When a sudden impact followed by stillness is detected — consistent with a fall — the app triggers the SOS screen automatically.

7.3 SOS Screen

FALL DETECTED screen appears	A full-screen red alert is shown immediately.
10-second countdown begins	A large countdown timer gives you time to cancel if it was a false alarm.
'I'M OKAY — Cancel SOS'	Tap this button to dismiss the alert within the countdown window.
If countdown reaches 0	VitalSync automatically calls the first emergency contact and sends an SMS with your current GPS coordinates.

7.4 Manual SOS

Tap the red SOS button on the Home Dashboard at any time to manually trigger the emergency countdown. Use this if you feel unwell and need immediate assistance.

8. App Settings & Permissions

Permission	Required For	Optional?
ACTIVITY_RECOGNITION	Step counter	No
ACCESS_FINE_LOCATION	Walk Tracker, SOS GPS	Yes*
BODY_SENSORS	Fall detection accelerometer	No
POST_NOTIFICATIONS	Rest timer, reminders	Yes
FOREGROUND_SERVICE	Background step counting	No
USE_BIOMETRIC	App lock (fingerprint/face)	Yes
CALL_PHONE	SOS emergency call	Yes*

* Denotes permissions required only for specific features (SOS / Walk Tracker). The app functions for other modules without these.

9. Troubleshooting

Q: Steps not counting	A: Ensure ACTIVITY_RECOGNITION permission is granted. Check Settings → Apps → VitalSync → Permissions. Some manufacturers require disabling battery optimization for background step counting.
Q: Walk Tracker map not loading	A: Ensure internet connection is active for map tile download. Verify ACCESS_FINE_LOCATION is granted. GPS accuracy improves outdoors.
Q: Fall detection triggered falsely	A: Vigorous movements like jumping or dropping the phone may trigger detection. Tap 'I'M OKAY' within 10 seconds to cancel.
Q: SOS call not made	A: Ensure CALL_PHONE permission is granted and at least one emergency contact with a valid phone number is saved.

Q: App crashes on launch	A: Try clearing app data: Settings → Apps → VitalSync → Clear Data. Reinstall if the issue persists.
Q: BMI not saving	A: Enter both weight and height before navigating away from the Health screen.

10. Support & Contact

VitalSync is an open-source project. For bug reports, feature requests, or questions, please use the GitHub Issues tab.

GitHub Repository	github.com/sohiaking/VitalSync
Issues & Bug Reports	github.com/sohiaking/VitalSync/issues
Developer	sohiaking
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